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Sliding bushing

Abstract

Provided is a sliding bushing in which an inner shaft member and an outer tube member are connected by a main rubber elastic body and sliding of the inner shaft member with respect to the main rubber elastic body is allowed. The inner shaft member includes a bulge part of a large diameter provided midway in an axial direction. The outer tube member includes a tapered part whose diameter decreases axially outward at both axial end portions. An outer peripheral surface of the bulge part in the inner shaft member includes a non-adhesive part that is non-adhesive to the main rubber elastic body and is allowed to slide with respect to the main rubber elastic body, and an outer peripheral surface axially outside the non-adhesive part in the inner shaft member includes an adhesive part to which the main rubber elastic body is adhered by vulcanization.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of PCT International Application No. PCT/JP2022/003275, filed on Jan. 28, 2022, which claims priority under 35 U.S.C § 119(a) to Japanese Patent Application No. 2021-059569, filed on Mar. 31, 2021. Each of the above application(s) is hereby expressly incorporated by reference, in its entirety, into the present application.

BACKGROUND

Technical Field

(1) The disclosure relates to a bushing used for, for example, a suspension bushing for an automobile or the like, and particularly relates to a sliding bushing in which sliding of an inner shaft member with respect to a main rubber elastic body is allowed.

Related Art

(2) Conventionally, there is known a bushing used for a suspension bushing of an automobile or the like, as disclosed in, for example, Japanese Patent Laid-Open No. 2010-159860 (Patent Document 1). The bushing of Patent Document 1 has a structure in which a shaft member and an outer cylinder are connected by a rubber-like elastic body.

(3) If both high spring properties in an axis-perpendicular direction and low spring properties in a torsional direction are required as in a suspension bushing, as shown in Patent Document 1, a structure may be adopted in which a first bulging part and a second bulging part are arranged concentrically by providing the first bulging part in the shaft member and providing the second bulging part in the outer cylinder.

(4) In the suspension bushing or the like, there are also cases where low spring properties are required for an input in the torsional direction. In this case, by making the shaft member non-adhesive to the rubber-like elastic body, relative rotation between the shaft member and the outer cylinder is allowed, and the required low spring properties in the torsional direction can be realized.

(5) However, as a result of a study by the present inventors, the following finding has been obtained. In the above-mentioned structure in Patent Document 1, when the shaft member is simply non-adhesive to the rubber-like elastic body and torsional displacement (rotation) is allowed, the rubber-like elastic body compressed between the first bulging part and the outer cylinder is likely to be deformed axially outward, and it becomes difficult to set high spring properties in the axis-perpendicular direction due to axially outward escape of the rubber-like elastic body. If relatively hard spring properties are required in the axis-perpendicular direction, there is a risk that the required properties may not be able to be satisfied.

SUMMARY

(6) According to one aspect of the disclosure, a sliding bushing is provided which has a structure in which an inner shaft member and an outer tube member are connected by a main rubber elastic body and sliding of the inner shaft member with respect to the main rubber elastic body is allowed. The inner shaft member includes a bulge part of a large diameter provided midway in an axial direction. The outer tube member includes a tapered part whose diameter decreases axially outward at both axial end portions. An outer peripheral surface of the bulge part in the inner shaft member includes a non-adhesive part that is non-adhesive to the main rubber elastic body and is allowed to slide with respect to the main rubber elastic body, and an outer peripheral surface axially outside the non-adhesive part in the inner shaft member includes an adhesive part to which the main rubber elastic body is adhered by vulcanization.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a longitudinal sectional view showing a suspension bushing as a first embodiment of the disclosure, corresponding to an I-I section of FIG. **2**.

(2) FIG. **2** is a front view of the suspension bushing shown in FIG. **1**.

(3) FIG. **3** is a longitudinal sectional view showing a portion of a suspension bushing as a second

embodiment of the disclosure.

DESCRIPTION OF THE EMBODIMENTS

(4) The disclosure provides a sliding bushing in which both hard spring properties in an axis-perpendicular direction and soft spring properties in a torsional direction and a prying direction can be advantageously set.

(5) Described below are aspects for understanding of the disclosure. However, the aspects described below are exemplary and may be adopted in combination with each other as appropriate. Moreover, components described in each aspect may be recognized and adopted independently wherever possible, and may be adopted in combination with any component described in another aspect as appropriate. Accordingly, in the disclosure, various different aspects may be realized and the disclosure is not limited to the aspects described below.

(6) According to one aspect, a sliding bushing has a structure in which an inner shaft member and an outer tube member are connected by a main rubber elastic body and sliding of the inner shaft member with respect to the main rubber elastic body is allowed. The inner shaft member includes a bulge part of a large diameter provided midway in an axial direction. The outer tube member includes a tapered part whose diameter decreases axially outward at both axial end portions. An outer peripheral surface of the bulge part in the inner shaft member includes a non-adhesive part that is non-adhesive to the main rubber elastic body and is allowed to slide with respect to the main rubber elastic body, and an outer peripheral surface axially outside the non-adhesive part in the inner shaft member includes an adhesive part to which the main rubber elastic body is adhered by vulcanization.

(7) According to the sliding bushing having the structure in accordance with this aspect, the outer peripheral surface of the bulge part of the inner shaft member includes the non-adhesive part to which the main rubber elastic body is not adhered. By providing the non-adhesive part on the outer peripheral surface of the bulge part, sliding of the inner shaft member and the main rubber elastic body in the bulge part is allowed, and low spring properties in a torsional direction and a prying direction may be achieved.

(8) The adhesive part to which the main rubber elastic body is adhered by vulcanization is provided axially outside the non-adhesive part on the outer peripheral surface of the inner shaft member. By providing the adhesive part axially outside the non-adhesive part, the main rubber elastic body is restrained by the inner shaft member at the adhesive part. When there is an input in an axis-perpendicular direction, axially outward deformation of the main rubber elastic body is restricted at the adhesive part. Hence, a high spring constant due to compression of the main rubber elastic body can be set in the axis-perpendicular direction. Further, by providing the tapered part at both axial end portions of the outer tube member, axially outward deformation of an outer peripheral portion of the main rubber elastic body is restricted by the tapered part. In this way, when the main rubber elastic body is compressed in the axis-perpendicular direction, in an inner peripheral portion of the main rubber elastic body, the main rubber elastic body that is likely to escape axially outward due to provision of the non-adhesive part is restrained by the adhesive part; in the outer peripheral portion of the main rubber elastic body, axially outward escape of the main rubber elastic body is restricted by the tapered part of the outer tube member. Accordingly, while the low spring properties in the prying direction and the torsional direction can be achieved due to provision of the non-adhesive part, the high spring properties in the axis-perpendicular direction can be realized.

(9) According to another aspect, in the sliding bushing described in the above aspect, in the main rubber elastic body, a minimum thickness dimension in a radial direction of a portion located on an outer periphery of the non-adhesive part is smaller than a maximum thickness dimension in the radial direction of a portion located on an outer periphery of the adhesive part.

(10) According to the sliding bushing having the structure in accordance with this aspect, by increasing the thickness dimension in the radial direction of the main rubber elastic body on the outer periphery of the adhesive part, a large free length of the main rubber elastic body can be

ensured in a portion where it is difficult to secure durability due to the restraint by adhesion, and the durability can be improved. Meanwhile, in the main rubber elastic body located on the outer periphery of the non-adhesive part where durability is less likely to be a problem, the thickness dimension in the radial direction is reduced. When there is a vibration input in the axis-perpendicular direction, hard spring properties due to compression of the non-adhesive part are exhibited.
(11) According to another aspect, in the sliding bushing described in any of the above aspects, a minimum inner diameter of the tapered part of the outer tube member is smaller than a maximum outer diameter of the bulge part of the inner shaft member.
(12) According to the sliding bushing having the structure in accordance with this aspect, coming off of the inner shaft member in the axial direction with respect to the outer tube member is restricted by the bulge part and the tapered part. In addition, for example, since the main rubber elastic body is compressed between the bulge part and the tapered part when there is an input in the axial direction, hard spring properties due to a compression spring component can be achieved in the axial direction.
(13) According to another aspect, in the sliding bushing described in any of the above aspects, a sliding layer having low friction is provided between overlapping surfaces of the non-adhesive part of the bulge part and the main rubber elastic body.
(14) According to the sliding bushing having the structure in accordance with this aspect, since sliding of the non-adhesive part of the inner shaft member and the main rubber elastic body is allowed relatively advantageously, the low spring properties in the prying direction and the torsional direction can further be achieved.
(15) According to another aspect, in the sliding bushing described in any of the above aspects, a recess opening on an outer peripheral surface is provided axially outside the bulge part in the inner shaft member, and the adhesive part is configured to include an inner surface of the recess.
(16) According to the sliding bushing having the structure in accordance with this aspect, a free length of the adhesive part of the main rubber elastic body can be increased, and the durability of the adhesive part may be improved.
(17) According to the disclosure, both hard spring properties in the axis-perpendicular direction and soft spring properties in the torsional direction and the prying direction can be advantageously set.
(18) Embodiments of the disclosure will be described below with reference to the drawings.
(19) FIG. **1** and FIG. **2** show a suspension bushing **10** for an automobile as a first embodiment of a sliding bushing having a structure in accordance with the disclosure. The suspension bushing **10** has a structure in which an inner shaft member **12** and an outer tube member **14** are elastically connected by a main rubber elastic body **16**.
(20) The inner shaft member **12** is, for example, a high-rigidity member made of metal, and has a cylindrical shape with a small diameter as a whole. The inner shaft member **12** may have a solid rod shape or the like. In that case, a fixing structure for fixation to a suspension arm or the like may be provided, for example, at both axial ends.
(21) The inner shaft member **12** is composed of a bulge part **18** in which an axial central portion protrudes toward an outer periphery and a diameter thereof is relatively large. The bulge part **18** has an outer peripheral surface having a substantially spherical annular shape convex toward the outer periphery. In the bulge part **18** of the present embodiment, an inner peripheral surface is a curved surface corresponding to the outer peripheral surface, and a thickness dimension is substantially constant over the entire axial direction. However, in the bulge part, the thickness dimension may also vary in the axial direction, and the inner peripheral surface may be, for example, a cylindrical surface extending straight in the axial direction.
(22) The inner shaft member **12** has small diameter parts **20** and **20** provided axially outside the bulge part **18**. The small diameter part **20** has a smaller outer diameter than the bulge part **18**, and extends axially outward from an axial end of the bulge part **18**. At an axial end of each small

diameter part **20** opposite to the bulge part **18**, a protrusion **22** protruding toward the outer periphery is provided over the entire periphery. In the small diameter part **20** of the inner shaft member **12**, a recess **24** opening on an outer peripheral surface is provided over the entire periphery between the protrusion **22** and the bulge part **18** in the axial direction.
(23) The outer tube member **14** is a high-rigidity member like the inner shaft member **12**. Compared to the inner shaft member **12**, the outer tube member **14** has a substantially cylindrical shape with a smaller thickness and a larger diameter, and has a shorter axial length dimension. In the outer tube member **14**, an axial central portion is a cylindrical part **26** extending linearly in the axial direction, and tapered parts **28** and **28** inclined axially outward toward the inner periphery are provided on both axial sides of the cylindrical part **26**. The tapered part **28** may be inclined at a constant angle with respect to the axial direction, and may also have an inclination angle varying in the axial direction. In the present embodiment, the inclination angle of the tapered part **28** with respect to the axial direction becomes smaller axially outward. The tapered part **28** may be provided in advance at the time of formation of the outer tube member **14**, and may also be formed, for example, in association with later-described diameter reduction processing of the outer tube member **14**.
(24) By providing the tapered parts **28** and **28** on both axial sides of the cylindrical part **26**, the outer tube member **14** has a sectional shape concave toward the inner periphery as a whole in the longitudinal section shown in FIG. **1**. In the outer tube member **14**, an axial dimension of the cylindrical part **26** is smaller than an axial length dimension of the bulge part **18**, and an axial length dimension of the entire outer tube member **14** is greater than the axial length dimension of the bulge part **18**.
(25) The inner shaft member **12** is inserted through the inner periphery of the outer tube member **14**, and the inner shaft member **12** and the outer tube member **14** are arranged concentrically. The inner shaft member **12** protrudes toward both axial sides with respect to the outer tube member **14**. Both axial ends of the outer tube member **14** are located on both axially outer sides of the bulge part **18** of the inner shaft member **12**. The bulge part **18** located inside the outer tube member **14** in the axial direction and the radial direction is arranged so as to be wrapped by the outer tube member **14** with a predetermined distance therebetween. A distance in the radial direction between the inner shaft member **12** and the outer tube member **14** is minimum in an axial center where the bulge part **18** and the cylindrical part **26** face each other, and gradually increases axially outward in the bulge part **18**. A minimum inner diameter R**1** of the tapered part **28** is smaller than a maximum outer diameter R**2** of the bulge part **18**. More preferably, a minimum inner diameter R**3** of the tapered part **28** at an axially outer end of a portion to which the main rubber elastic body **16** is fixed is smaller than the maximum outer diameter R**2** of the bulge part **18**, and the main rubber elastic body **16** is continuously provided in the axial direction between axially facing surfaces of the bulge part **18** and the tapered part **28**.
(26) The inner shaft member **12** and the outer tube member **14** are elastically connected by the main rubber elastic body **16**. The main rubber elastic body **16** has a cylindrical shape as a whole, and is provided to connect facing surfaces of an outer peripheral surface of the inner shaft member **12** and an inner peripheral surface of the outer tube member **14**. The main rubber elastic body **16** is formed as an integrally vulcanized molded product including the inner shaft member **12** and the outer tube member **14**. In the main rubber elastic body **16**, in a portion arranged so as to directly fill a space between radially facing surfaces of the inner shaft member **12** and the outer tube member **14**, a radial thickness dimension at both axial end portions is greater than that in the axial center. However, the main rubber elastic body **16** may also extend in the axial direction with a substantially constant radial thickness dimension.
(27) By performing diameter reduction processing such as drawing on the outer tube member **14** after vulcanization molding of the main rubber elastic body **16**, tensile stress due to thermal contraction acting on the main rubber elastic body **16** is reduced, and durability of the main rubber

elastic body **16** may be improved. At the time of diameter reduction processing of the outer tube member **14**, by further reducing the diameter of an axial end of the outer tube member **14**, the tapered parts **28** and **28** are formed in the outer tube member **14**. Accordingly, while the inner shaft member **12** can be inserted through the outer tube member **14** before molding of the main rubber elastic body **16**, the minimum inner diameter R**1** of the tapered parts **28** and **28** can be made smaller than the maximum outer diameter R**2** of the bulge part **18** after molding of the main rubber elastic body **16**. Coming off of the inner shaft member **12** with respect to the outer tube member **14** is prevented by indirect engagement between the bulge part **18** and the tapered parts **28** and **28** via the main rubber elastic body **16**. By making the minimum inner diameter R**3** of the portion in the tapered parts **28** and **28** where the main rubber elastic body **16** is fixed smaller than the maximum outer diameter R**2** of the bulge part **18**, it is possible to achieve hard spring properties due to compression of the main rubber elastic body **16** between the bulge part **18** and the tapered parts **28** and **28** when there is a vibration input in the axial direction. In FIG. **1**, the outer tube member **14** before the diameter reduction processing is indicated by a two-dot chain line.
(28) The outer tube member **14** is adhered to the main rubber elastic body **16** by vulcanization. Between overlapping surfaces of the outer tube member **14** and the main rubber elastic body **16**, no sliding layer like that between overlapping surfaces of the inner shaft member **12** and the main rubber elastic body **16** is provided. Hence, when the outer tube member **14** is subjected to diameter reduction, problems such as wrinkles in the sliding layer are unlikely to occur.
(29) The main rubber elastic body **16** is arranged in an axial region straddling the outer peripheral surface of the bulge part **18** and the outer peripheral surface of the small diameter parts **20** and **20** in the inner shaft member **12**. In the main rubber elastic body **16**, both axial end portions are adhered by vulcanization to adhesive parts **30** and **30** composed of the outer peripheral surfaces on both axial sides of the inner shaft member **12**, and an axial central portion non-adhesively overlaps a non-adhesive part **32** composed of an outer peripheral surface of an axial central portion of the inner shaft member **12**. In the main rubber elastic body **16**, an axial length dimension of the portion arranged so as to directly fill the space between the radially facing surfaces of the inner shaft member **12** and the outer tube member **14** may be a length limited to being on the outer peripheral surface of the bulge part **18** of the inner shaft member **12**, or may be an axial length extending from the bulge part **18** to the small diameter parts **20** and **20** extending on both axial sides.
(30) The adhesive parts **30** and **30** are composed of the outer peripheral surface of the axial end of the bulge part **18** and the outer peripheral surface of the small diameter parts **20** and **20** constituting both axial end portions of the inner shaft member **12**. In the present embodiment, the adhesive parts **30** and **30** are configured to include an inner surface of the recesses **24** and **24** of the small diameter parts **20** and **20**. The adhesive parts **30** and **30** extend axially outward from the axial end of the outer tube member **14** and reach an outer peripheral surface of the protrusions **22** and **22**. An inner peripheral portion of the main rubber elastic body **16** includes an inner fixing part **34** fixed to the inner surface of the recess **24**. An axially inner end of the inner fixing part **34** is located axially inwardly offset from a hollow part **38**. An axially outer portion of the inner fixing part **34** is located axially outside a deepest portion of the hollow part **38**, and an outer peripheral surface of the axially outer portion of the inner fixing part **34** is exposed. It is desirable that the adhesive part **30** include the outer peripheral surface of the small diameter parts **20** and **20** protruding toward both axial sides of the bulge part **18**.
(31) In the present embodiment, the non-adhesive part **32** is composed of the outer peripheral surface of the axial central portion of the bulge part **18**. Since the inner shaft member **12** and the main rubber elastic body **16** are non-adhesive at the non-adhesive part **32**, sliding (relative displacement of the overlapping surfaces) of the inner shaft member **12** and the main rubber elastic body **16** is allowed at the non-adhesive part **32**. In the present embodiment, the non-adhesive part **32** is covered with a sliding layer **36** made of a low-friction coating material, and sliding of the inner shaft member **12** and the main rubber elastic body **16** is likely to occur at the non-adhesive

part **32**. The sliding layer **36** is formed by a known method such as coating or vapor deposition using, for example, a known sliding coating material such as a fluororesin or molybdenum disulfide. It is desirable that the non-adhesive part **32** include the center of the bulge part **18**. For example, the non-adhesive part **32** may be set to extend over a region extending from a maximum outer diameter part of the bulge part **18** toward both axial sides and axially outward of a point reaching ½ of a difference between the maximum outer diameter and a minimum outer diameter of the bulge part **18**.
(32) The outer peripheral surface of the main rubber elastic body **16** is adhered by vulcanization to the inner peripheral surface of the outer tube member **14**. The main rubber elastic body **16** is fixed over an inner peripheral surface of the cylindrical part **26** and the inner peripheral surface of the tapered parts **28** and **28** in the outer tube member **14**.
(33) In the main rubber elastic body **16**, a minimum thickness dimension W**1** in the radial direction of a portion located on an outer periphery of the non-adhesive part **32** is smaller than a maximum thickness dimension W**2** in the radial direction of a portion located on an outer periphery of the adhesive part **30**. The minimum thickness dimension W**1** in the radial direction of the portion located on the outer periphery of the non-adhesive part **32** is smaller than a minimum thickness dimension W**3** in the radial direction of a portion in the adhesive part **30** that continuously connects the inner shaft member **12** and the outer tube member **14** in the radial direction. On an axial end face of the main rubber elastic body **16**, hollow parts **38** and **38** having a concave shape opening in the axial direction are formed over the entire periphery between the inner shaft member **12** and the outer tube member **14** in the radial direction, and a bottom of the hollow parts **38** and **38** reaches the outer periphery of the bulge part **18**. Depth, size, shape or the like of the hollow part **38** is not particularly limited. For example, the hollow part **38** may have a depth not reaching the outer periphery of the bulge part **18** from the axial end face of the main rubber elastic body **16**. The hollow part **38** may not be necessarily provided.
(34) The suspension bushing **10** having such a structure connects a vehicle body and a suspension arm in a vibration-proof manner by attachment of the inner shaft member **12** to the vehicle body side of a subframe (not shown) or the like and attachment of the outer tube member **14** to the suspension arm side (not shown).
(35) When a vibration in the axis-perpendicular direction (radial direction) is input between the inner shaft member **12** and the outer tube member **14**, since the main rubber elastic body **16** is compressed in the axis-perpendicular direction between the inner shaft member **12** and the outer tube member **14**, hard spring properties due to a compression spring component are exhibited. In particular, since the bulge part **18** is provided in the inner shaft member **12**, and the radial thickness dimension of the main rubber elastic body **16** is reduced on an outer peripheral side of the bulge part **18**, hard spring properties can be achieved by compression of the main rubber elastic body **16**.
(36) Since the main rubber elastic body **16** is slidable with respect to the non-adhesive part **32** of the inner shaft member **12** including the outer peripheral surface of the bulge part **18**, when compressed in the axis-perpendicular direction, the main rubber elastic body **16** tends to be deformed axially outward according to the shape of the outer peripheral surface of the bulge part **18**. On both axial sides of the non-adhesive part **32**, the adhesive parts **30** and **30** to which the main rubber elastic body **16** is adhered by vulcanization are provided, and the deformation of the main rubber elastic body **16** is restricted at the adhesive parts **30** and **30**. Accordingly, the main rubber elastic body **16** located on the outer periphery of the non-adhesive part **32** becomes less likely to be deformed axially outward. By preventing the main rubber elastic body **16** from escaping axially outward, hard spring properties due to the compression in the axis-perpendicular direction are effectively exhibited.
(37) Further, by providing the tapered parts **28** and **28** at both axial ends of the outer tube member **14**, axially outward escape of an outer peripheral portion of the main rubber elastic body **16** is suppressed by the tapered parts **28** and **28**. Accordingly, hard spring properties in the axis-

perpendicular direction can be relatively effectively set.
(38) In the main rubber elastic body **16**, a free length of a portion connecting the adhesive part **30** of the inner shaft member **12** and the outer tube member **14** is greater than a free length of a portion connecting the bulge part **18** and the outer tube member **14**. Accordingly, the durability of the main rubber elastic body **16** may be improved in the portions where damage is likely to be a problem due to the adhesion to both the inner shaft member **12** and the outer tube member **14**.
(39) In the present embodiment, the adhesive part **30** of the inner shaft member **12** is configured to include the inner surface of the recesses **24** and **24**, and the free length of the portion in the main rubber elastic body **16** that connects the adhesive part **30** and the outer tube member **14** is ensured to be relatively large by the recess **24**. Accordingly, the durability of the portion in the main rubber elastic body **16** that connects the adhesive part **30** of the inner shaft member **12** and the outer tube member **14** may be improved.
(40) When a vibration in the prying direction is input between the inner shaft member **12** and the outer tube member **14**, since the main rubber elastic body **16** undergoes shear deformation between the bulge part **18** of the inner shaft member **12** and the outer tube member **14** and the compression spring component is reduced, low spring properties are realized in the prying direction. Since the inner shaft member **12** and the main rubber elastic body **16** are non-adhesive and slidable at the non-adhesive part **32** set on the outer peripheral surface of the bulge part **18**, a shear spring component is also reduced and a spring constant in the prying direction is further reduced. By providing the hollow part **38** in an axially outer portion of the main rubber elastic body **16** located between the adhesive part **30** and the outer tube member **14**, the compression spring component of the main rubber elastic body **16** is further reduced during prying displacement of the inner shaft member **12** and the outer tube member **14**.
(41) When a vibration in the torsional direction is input between the inner shaft member **12** and the outer tube member **14**, since the main rubber elastic body **16** undergoes shear deformation between the bulge part **18** of the inner shaft member **12** and the outer tube member **14** and the compression spring component is reduced, low spring properties are realized in the torsional direction. Since the inner shaft member **12** and the main rubber elastic body **16** are non-adhesive and slidable at the non-adhesive part **32** set on the outer peripheral surface of the bulge part **18**, a shear spring component is also reduced and a spring constant in the torsional direction is further reduced.
(42) In the present embodiment, the sliding layer **36** having low friction is provided in the non-adhesive part **32** of the inner shaft member **12**, and frictional resistance during sliding between the non-adhesive part **32** and the inner peripheral surface of the main rubber elastic body **16** is reduced. Accordingly, low spring properties in the prying direction and the torsional direction can be relatively effectively realized.
(43) FIG. **3** shows a portion of a suspension bushing **40** as a second embodiment of the disclosure. The suspension bushing **40** includes an outer tube member **42**, and has a structure in which a relief part **44** that curves toward the outer periphery is provided axially outside the tapered part **28** of the outer tube member **42**. In the following description, the members and portions substantially the same as those of the first embodiment are designated by the same reference numerals in the drawings, and the description thereof will be omitted.
(44) The relief part **44** is inclined axially outward so as to be located on the outer periphery. The relief part **44** of the present embodiment has a curved sectional shape in which an inclination angle with respect to the axial direction increases axially outward. By providing the relief part **44**, an end face of the outer tube member **42** faces the outer peripheral side, and an inner peripheral surface **46** of an axial end of the outer tube member **42** is set as a smooth curved surface without edges. The outer tube member **42** faces the inner fixing part **34** of the main rubber elastic body **16** in the radial direction at the curved inner peripheral surface **46** composed of an end of the tapered part **28** and the relief part **44**.
(45) According to the suspension bushing **40** according to the present embodiment like this, an

edge of the axial end of the outer tube member **42** can be prevented from coming into contact with the main rubber elastic body **16** when there is a vibration input, and damage to the main rubber elastic body **16** is avoided. By providing the relief part **44** having a curved section at the axial end of the outer tube member **42**, deformation rigidity of the outer tube member **42** may be improved.
(46) Although the embodiments of the disclosure have been described in detail above, the disclosure is not limited by the specific description thereof. For example, the adhesive part **30** of the inner shaft member **12** may be a portion of the outer peripheral surface of the small diameter part **20**, or may be the entire outer peripheral surface of the small diameter part **20**, or may include a portion of the outer peripheral surface of the bulge part **18** in addition to the entire outer peripheral surface of the small diameter part **20**. The non-adhesive part **32** of the inner shaft member **12** may be a portion of the outer peripheral surface of the bulge part **18**, or may be the entire outer peripheral surface of the bulge part **18**, or may include a portion of the outer peripheral surface of the small diameter part **20** in addition to the entire outer peripheral surface of the bulge part **18**.
(47) It suffices if the bulge part **18** of the inner shaft member **12** is provided midway in the axial direction of the inner shaft member **12**. For example, the bulge part **18** of the inner shaft member **12** may deviate toward either side in the axial direction with respect to an axial center of the inner shaft member **12**. It is desirable that the bulge part **18** have an outer peripheral surface shape that is an arc shape in the longitudinal section corresponding to FIG. **1**. However, an outer peripheral surface shape that is, for example, a substantially trapezoidal shape, may also be adopted. The recess **24** of the inner shaft member **12** is not essential. The recess **24** may be opened axially outward, for example, with the protrusion **22** omitted, and both ends of the inner shaft member **12** may accordingly have a small diameter up to an end edge.
(48) In the tapered parts **28** and **28** of the outer tube member **14**, the inner peripheral surface and the outer peripheral surface may have different inclination angles from each other. For example, in the tapered part, it may be that only the inner peripheral surface is inclined axially outward toward the inner periphery, or is thickened axially outward. The maximum outer diameter R**2** of the bulge part **18** of the inner shaft member **12** may be smaller than the minimum inner diameter R**1** of the tapered parts **28** and **28** of the outer tube member **14**.
(49) For purposes of avoiding interference of the outer tube member **14** at the time of prying input or the like, in the inner shaft member **12** and/or the inner fixing part **34** of the main rubber elastic body **16**, a recess may be provided opening in a portion facing an end of the outer tube member **14** and extending in a circumferential direction.
(50) A structure coated with a coating material has been illustrated as an example of the sliding layer **36**. However, a sliding layer may also be composed of a sliding liner that is a braided body made of low-friction fibers and is arranged to overlap the non-adhesive part **32** of the inner shaft member **12**. The sliding layer is not essential and may be partially provided in at least a portion of the non-adhesive part **32** instead of the entire non-adhesive part **32**.
(51) It is possible that the material for forming the main rubber elastic body **16** includes a self-lubricating rubber material in which a surface friction coefficient is reduced by mixing with oil or the like. Accordingly, slidability between the inner shaft member **12** and the main rubber elastic body **16** can further be improved.
(52) The adhesive part **30** may be adhered over the entire surface, and may also be adhered, for example, at predetermined intervals in the circumferential direction, or may be partially adhered in a plurality of regions. Alternatively, in a boundary region between the adhesive part **30** and the non-adhesive part **32**, it is possible to provide a partially adhered region or a region where a ratio of adhesive area gradually varies in the axial direction and distribute or transfer a rubber binding force by adhesion. Similarly, the non-adhesive part **32** is not limited to the aspect in which the non-adhesive part **32** is non-adhesive over the entire surface in response to realization of the required spring properties or durability. As can be seen from this, the adhesive part and the non-adhesive

part in the disclosure may be interpreted as relative.
(53) In the above embodiment, an example has been shown in which the disclosure is applied to a suspension bushing for an automobile. However, the disclosure is also applicable to a sliding bushing other than a suspension bushing.

Claims

1. A sliding bushing, having a structure in which an inner shaft member and an outer tube member are connected by a main rubber elastic body and sliding of the inner shaft member with respect to the main rubber elastic body is allowed, wherein the inner shaft member comprises a bulge part of a large diameter provided midway in an axial direction; the outer tube member comprises a tapered part whose diameter decreases axially outward at both axial end portions; an outer peripheral surface of the bulge part in the inner shaft member comprises a non-adhesive part that is non-adhesive to the main rubber elastic body and is allowed to slide with respect to the main rubber elastic body, and an outer peripheral surface axially outside the non-adhesive part in the inner shaft member comprises an adhesive part to which the main rubber elastic body is adhered by vulcanization, wherein the adhesive part comprises a first adhesive part and a second adhesive part, and the non-adhesive part is disposed between the first adhesive part and the second adhesive part.
2. The sliding bushing according to claim 1, wherein, in the main rubber elastic body, a minimum thickness dimension in a radial direction of a portion located on an outer periphery of the non-adhesive part is smaller than a maximum thickness dimension in the radial direction of a portion located on an outer periphery of the adhesive part.
3. The sliding bushing according to claim 1, wherein a minimum inner diameter of the tapered part of the outer tube member is smaller than a maximum outer diameter of the bulge part of the inner shaft member.
4. The sliding bushing according to claim 1, wherein a sliding layer having low friction is provided between overlapping surfaces of the non-adhesive part of the bulge part and the main rubber elastic body.
5. The sliding bushing according to claim 1, wherein a recess opening on an outer peripheral surface is provided axially outside the bulge part in the inner shaft member, and the adhesive part is configured to comprise an inner surface of the recess.